

Encoding Categorical Risk Factors in IRB LGD and EAD Models

A Comparison of the Target-Based Encoding Methods

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Linear Regression in LGD and EAD Modeling

- The development of an Internal Ratings-Based (IRB) model encompasses two primary areas: the creation of the risk differentiation function and risk quantification.
- A common approach to developing the risk differentiation function in Loss Given Default (LGD) and Exposure at Default (EAD) models is linear regression, specifically Ordinary Least Squares (OLS). Although linear regression is usually considered part of multifactor analysis, the preceding steps of risk factor engineering should also be viewed as essential to properly applying linear regression in this context.
- The development of LGD and EAD risk differentiation functions typically involves both numeric and categorical risk factors.
- The final model typically comprises a set of categorical risk factors selected through OLS regression. Some of these are the result of discretizing numeric risk factors, while others are inherently categorical by nature.
- To prepare a categorical risk factor for linear regression, practitioners must select a specific encoding method.
- Commonly used encoding methods are those that use information from the target variable to encode the bins (modalities) of the risk factors. This is typically known as a target-based encoding process. As the selection of the encoding method can impact subsequent modeling steps, practitioners are encouraged to understand the main characteristics and possible implications of each method.

Target-Based Encoding Methods in Practice

- In simple terms, encoding refers to transforming categorical risk factors into one or more numeric variables suitable for use in linear regression models.
- Encoding is often treated as a mechanical step required to run the regression, with insufficient attention given to its alignment with preceding modeling steps and with the criteria used to select the final model within the linear regression framework.
- This presentation focuses on comparing the following four encoding methods, raising awareness of their interpretation and their alignment with other modeling steps:
 - 1 Weights of Evidence (WoE) encoding;
 - 2 mean target encoding;
 - 3 baseline-adjusted target encoding;
 - 4 James-Stein encoding.

Target-Based Encoding Methods in Practice cont.

Woe Encoding Method

In LGD and EAD IRB modeling, practitioners often draw comparisons with the WoE encoding used in Probability of Default (PD) modeling. Consequently, one of the commonly applied methods is WoE encoding adjusted for a continuous target. The following formula presents the calculation of WoE values for bin i of a categorical risk factor:

$$WoE_i = \ln \left(\frac{\bar{y}_i}{\bar{y}} \right)$$

where:

- $\bar{y}_i$ denotes the average value of the target for bin i ;
- $\bar{y}$ is the average value of the target for the full sample.

Mean Target encoding Method

Another commonly used encoding method is mean target encoding. In essence, this approach assigns to each bin of a risk factor the average value of the target within that bin. Following the notation from the formula above, mean target encoding assigns the value $\bar{y}_i$ to bin i of the analyzed risk factor.

Baseline-Adjusted Mean Target Encoding Method

The baseline-adjusted mean target encoding simply subtracts the overall sample mean of the target from the mean target value of each bin of the risk factor. The main difference between this method and mean target encoding lies in interpretability, where the baseline-adjusted method fully aligns with the interpretation of estimates from WoE encoding in PD models. Practitioners should note that predictions from regression using this method and mean target encoding produce identical results.

Target-Based Encoding Methods in Practice cont.

James-Stein Encoding Method

The main characteristic of James–Stein (JS) encoding is that it “shrinks” the category mean toward the overall mean based on weights assigned to that specific bin and the overall mean. The weights can be calculated simply using the number of observations or by incorporating a measure of variability. The following formula presents the JS encoding method, which uses the standard error of the mean to determine the weights:

$$y_i^{\text{JS}} = (1 - B_i) \cdot \bar{y}_i + B_i \cdot \bar{y}$$
$$B_i = \frac{se_i^2}{se_i^2 + se^2}$$

where:

- se_i is the standard error of the mean of the target variable for bin i calculated as $\frac{\sigma_i}{n_i}$ with n_i being the number of observations in bin i ;
- se^2 is the standard error of the mean of the target variable in the overall sample, calculated as $\frac{\sigma}{n}$ with n being the overall sample size.

Simulation Study

The purpose of this simulation study is to demonstrate the differences and similarities in linear regression output when applying different encoding methods. In addition to differences in the technical output, practitioners should be aware of the varying interpretations of the estimated coefficients associated with risk factors or dummy variables, as well as the interpretation of the p-value when it is used as one of the criteria for selecting the final model.

For full replicability, the simulation dataset is available at this [link](#).

The dataset consists of the model target variable, `lgd`, and three categorical risk factors whose modalities can be sorted in alphabetical order, reflecting a specific relationship with the target variable. The following table presents a summary of the number of observations, as well as the mean and variance of `lgd` for each modality of each risk factor.

##	rf	bin	no	lgd_average	lgd_var
## 1	rf1	01 (-Inf,20.1001)	530	0.4614	0.1586
## 2	rf1	02 [20.1001,Inf)	666	0.2969	0.1177
## 3	rf2	01 (-Inf,58.1375)	92	0.6285	0.1472
## 4	rf2	02 [58.1375,142.6312)	169	0.4341	0.1963
## 5	rf2	03 [142.6312,531.6402)	570	0.3594	0.1307
## 6	rf2	04 [531.6402,642.8232)	270	0.3077	0.1243
## 7	rf2	05 [642.8232,Inf)	95	0.2431	0.0692
## 8	rf3	01 (-Inf,-0.0417)	131	0.5438	0.1790
## 9	rf3	02 [-0.0417,-0.0208)	147	0.4346	0.1529
## 10	rf3	03 [-0.0208,0.0506)	708	0.3553	0.1346
## 11	rf3	04 [0.0506,Inf)	210	0.2647	0.1069

Simulation Results

Woe Encoding Method

Regression estimates:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	0.3868	0.0105	36.7090	0
## rf1	0.2190	0.0518	4.2235	0
## rf2	0.2848	0.0515	5.5315	0
## rf3	0.2322	0.0558	4.1620	0

Prediction sample:

	1	2	3	4	5	6	7	8
##	0.4179325	0.4179325	0.3213801	0.4179325	0.4179325	0.1417242	0.1417242	0.2530823

Mean Target encoding Method

Regression estimates:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-0.3279	0.0655	-5.0047	0
## rf1	0.5771	0.1391	4.1488	0
## rf2	0.7044	0.1249	5.6389	0
## rf3	0.6054	0.1449	4.1788	0

Prediction sample:

	1	2	3	4	5	6	7	8
##	0.4065697	0.4065697	0.3116282	0.4065697	0.4065697	0.1748919	0.1748919	0.2568341

Simulation Results cont.

Baseline-adjusted Mean Target Encoding Method

Regression estimates:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	0.3698	0.0104	35.5095	0
## rf1	0.5771	0.1391	4.1488	0
## rf2	0.7044	0.1249	5.6389	0
## rf3	0.6054	0.1449	4.1788	0

Prediction sample:

	1	2	3	4	5	6	7	8
##	0.4065697	0.4065697	0.3116282	0.4065697	0.4065697	0.1748919	0.1748919	0.2568341

James-Stein Encoding Method

Regression estimates:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-4.3340	0.5153	-8.4099	0e+00
## rf1	1.7957	0.4120	4.3588	0e+00
## rf2	6.4812	1.2207	5.3094	0e+00
## rf3	4.5885	1.1451	4.0069	1e-04

Prediction sample:

	1	2	3	4	5	6	7	8
##	0.4214116	0.4214116	0.3218956	0.4214116	0.4214116	0.1638221	0.1638221	0.2562102

Discussion Points

- How does the selection of a target-based encoding method align with the risk factor binning process?
- How does the selection of a target-based encoding method impact the interpretation of the estimated coefficients?
- How does the selection of a target-based encoding method impact the analysis of risk factor importance in the final model?
- How does the selection of a target-based encoding method align with business expectations regarding estimated risk level changes between risk factor modalities?
- How does the selection of the encoding method impact the validation of the model design?